EC325 Concept Check 5

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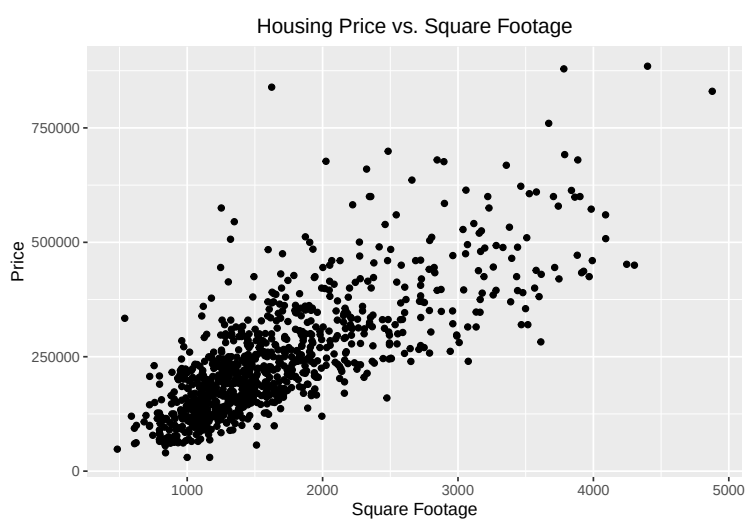
```
knitr::opts_chunk$set(fig.pos = "H", message = FALSE, warning = FALSE)
library(tidyverse)
library(modeldata)
data(Sacramento)
library(ggplot2)
```

Question 1: Visualizing the Relationship

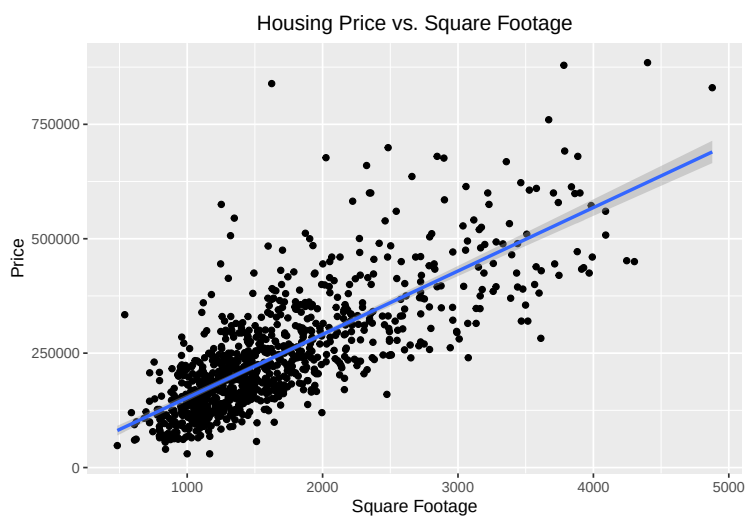
A) Square Footage vs. Price

```
ggplot(data = Sacramento, aes(x = sqft, y = price)) +
  geom_point() +
  xlab("Square Footage") +
  ylab("Price") +
  ggtitle("Housing Price vs. Square Footage") +
  theme(plot.title = element_text(hjust = 0.5))
```



B) Square Footage vs. Price with line of best fit

```
ggplot(data = Sacramento, aes(x = sqft, y = price)) +
  geom_point() +
  xlab("Square Footage") +
  ylab("Price") +
  ggtitle("Housing Price vs. Square Footage") +
  theme(plot.title = element_text(hjust = 0.5))+
  geom_smooth(method = "lm")
```



C) I expect $\hat{\beta}_1$ to be positive because the line of best fit has a positive slope.

Question 2: From Bivariate to Multivariate

A) Bivariate model

```
biv.model = lm(price ~ sqft, data = Sacramento)
summary(biv.model)
```

```
##
## Call:
## lm(formula = price ~ sqft, data = Sacramento)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -231889  -54717  -11822   38993  600141
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 13859.393   6948.714   1.995   0.0464 *
## sqft          138.546      3.796  36.495   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 84130 on 930 degrees of freedom
## Multiple R-squared:  0.5888, Adjusted R-squared:  0.5884
## F-statistic:  1332 on 1 and 930 DF,  p-value: < 2.2e-16
```

The coefficient on `sqft` is 138.55.

B) Multivariate model

```r
multi.model = lm(price ~ sqft + beds, data = Sacramento)
summary(multi.model)
```

```
##
## Call:
## lm(formula = price ~ sqft + beds, data = Sacramento)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -200938  -52196   -9148   37268  620281
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  60850.544  10424.530   5.837 7.33e-09 ***
## sqft           161.336      5.339  30.218  < 2e-16 ***
## beds        -26035.145   4366.612  -5.962 3.53e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 82610 on 929 degrees of freedom
## Multiple R-squared:  0.604,  Adjusted R-squared:  0.6031
## F-statistic: 708.5 on 2 and 929 DF,  p-value: < 2.2e-16
```

The coefficient on `sqft` is now 161.34 and the coefficient on `beds` is -26035.15.

C) The coefficient on `sqft` increased by over 16% after adding `beds` to the regression.

D) The coefficient on `sqft` likely changed after adding `beds` as a predictor variable because when holding square footage fixed, more bedrooms would imply smaller bedrooms and less space per person in a household. A 4 bedroom, 1500 square foot house is likely more cramped than a house of the same size with two bedrooms.

Question 3: Interpreting Coefficients and Adding Controls

A) Adding `baths` to the model.

```r
A3.model = lm(price ~ sqft + beds + baths, data = Sacramento)
summary(A3.model)
```

```
##
## Call:
## lm(formula = price ~ sqft + beds + baths, data = Sacramento)
```

```
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -202306  -52808   -9260   37341  610425
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  58481.747  10755.143   5.438 6.91e-08 ***
## sqft           158.167      6.403  24.702  < 2e-16 ***
## beds        -27035.285   4507.272  -5.998 2.86e-09 ***
## baths         5342.591   5958.426   0.897     0.37
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 82610 on 928 degrees of freedom
## Multiple R-squared:  0.6043, Adjusted R-squared:  0.6031
## F-statistic: 472.5 on 3 and 928 DF,  p-value: < 2.2e-16
```

B) Holding bedrooms and bathrooms constant, 1 square foot size increase raises a house's predicted sale price by $158.17.

C) When holding bedrooms and square footage constant, an an increase in bathroom count of 1 raises its predicted price by $5342.59. However, it is worth noting that when holding `sqft` and `beds` constant, `baths` is not a statistically significant predictor of a house's market value.

D) Tibble creation

```r
comparison = tibble(
  model = c("biv.model",
            "multi.model",
            "A3.model"),
  sqft_coef = c(coef(biv.model)["sqft"],
                coef(multi.model)["sqft"],
                coef(A3.model)["sqft"]))
comparison
```

```
## # A tibble: 3 x 2
##   model       sqft_coef
##   <chr>           <dbl>
## 1 biv.model        139.
## 2 multi.model      161.
## 3 A3.model         158.
```

E) Adding `baths` did not substantially change the the coefficient on `sqft` compared to the model with only `sqft` and `beds`. This suggests `baths` is not an important confounding variable for the square footage–price relationship.

Question 4: Omitted Variable Bias

Data Generation

```r
set.seed(325)
n <- 500

# True DGP: y = 10 + 3*x1 + 5*x2 + error
# x1 and x2 are correlated [cor(x1, x2) = 0.6]
x2 <- rnorm(n)
x1 <- 0.6 * x2 + rnorm(n)
y <- 10 + 3 * x1 + 5 * x2 + rnorm(n)

sim_data <- tibble(y, x1, x2)
```

A) The true value of β_1 is 3.

B) Bivariate model

```r
B5.model = lm(y ~ x1, data = sim_data)
summary(B5.model)
```

```
##
## Call:
## lm(formula = y ~ x1, data = sim_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.1064  -2.8844   0.3733   2.8626  10.7630
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   9.6585     0.1938   49.83   <2e-16 ***
## x1            4.7740     0.1760   27.13   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.333 on 498 degrees of freedom
## Multiple R-squared:  0.5964, Adjusted R-squared:  0.5956
## F-statistic: 735.9 on 1 and 498 DF,  p-value: < 2.2e-16
```

The bivariate regression coefficient is $\hat{\beta}_1 = 4.77$. This means it is biased because the true value of $\beta_1 = 3$. Since $4.77 > 3$, that means the model is biased upward.

C) Multivariate model

```r
C5.model = lm(y ~ x1 + x2, data = sim_data)
summary(C5.model)
```

```
##
## Call:
## lm(formula = y ~ x1 + x2, data = sim_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.12889 -0.69547  0.05836  0.66963  2.70192
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 10.07320    0.04354  231.37   <2e-16 ***
## x1           2.89858    0.04381   66.16   <2e-16 ***
## x2           5.04938    0.05189   97.31   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9687 on 497 degrees of freedom
## Multiple R-squared:  0.9799, Adjusted R-squared:  0.9798
## F-statistic: 1.21e+04 on 2 and 497 DF,  p-value: < 2.2e-16
```

The value of $\hat{\beta}_1$ for the multivariate model is 2.90. While it is still slightly biased (this time downward), it is much closer to the true value.

D) Bias estimation

```r
D5.model = lm(x2 ~ x1, data = sim_data)
summary(D5.model)
```

```
##
## Call:
## lm(formula = x2 ~ x1, data = sim_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.14847 -0.54694  0.03687  0.54697  2.17317
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.08214    0.03742  -2.195   0.0286 *
## x1           0.37141    0.03397  10.933   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8365 on 498 degrees of freedom
## Multiple R-squared:  0.1935, Adjusted R-squared:  0.1919
## F-statistic: 119.5 on 1 and 498 DF,  p-value: < 2.2e-16
```

The coefficient for `x1`, $\tilde{\delta}_1$ is 0.371. Using the OVB formula, $\text{Bias}(\hat{\beta}_1) = \beta_2\tilde{\delta}_1$. We know $\beta_2 = 5$, so bias is then equal to $5 \cdot 0.371 = 1.86$. The predicted bias is positive. Omitted variable bias has upward bias on the coefficient.

E) The true bias is $\tilde{\beta}_1 - \hat{\beta}_1 = 4.777 - 3 = 1.77$. This value is very close to the predicted bias, proving the efficacy of the OVB formula for estimating direction and magnitude of the bias.